

Lecture #28

Content to be covered

- ❖ Potential Barrier
 - ❑ Quantum Tunnelling
 - ❑ Tunnelling Probability
 - ❑ Numerical Problems

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

Quantum Tunnelling

Penetration of particles through rectangular potential barrier

Potential Barrier: when a particle approaches a region in which the entrance of the particle is opposed by some force, then the region is said to form a barrier for the particle.

Tunnel effect: when a particle is able to cross a potential barrier even when its energy is less than the barrier height, then this phenomenon is called “tunnel effect”. It is purely quantum mechanical phenomenon, never realizable classically.

The emission of α -particles from atomic nuclei is an example of tunnel effect.

Tunnel Diode, Scanning tunneling Microscope

Potential Barrier

If E is the energy of the particle then according to

Classical theory:

If $E > V_0$ particle will always cross the barrier

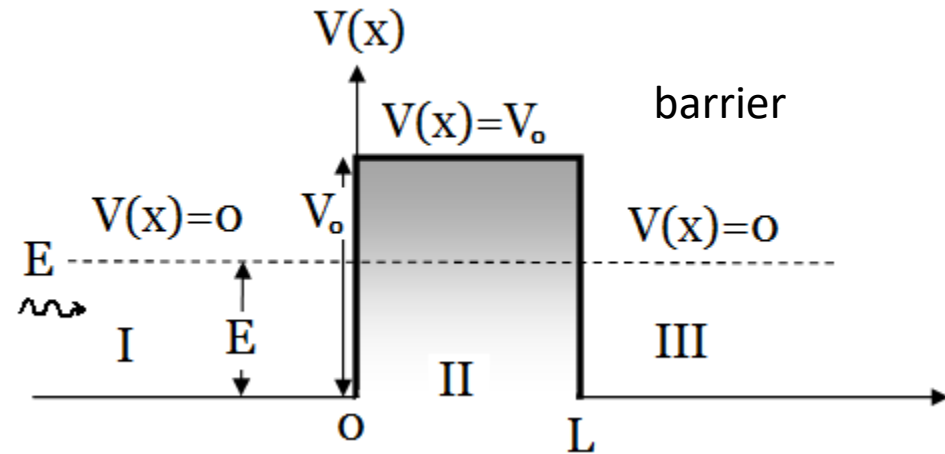
If $E < V_0$ Particle will always be reflected back
it can never penetrate through the barrier.

But, according to Quantum Mechanics:

when $E > V_0$, there is a finite probability that the particle will be reflected back from the barrier.

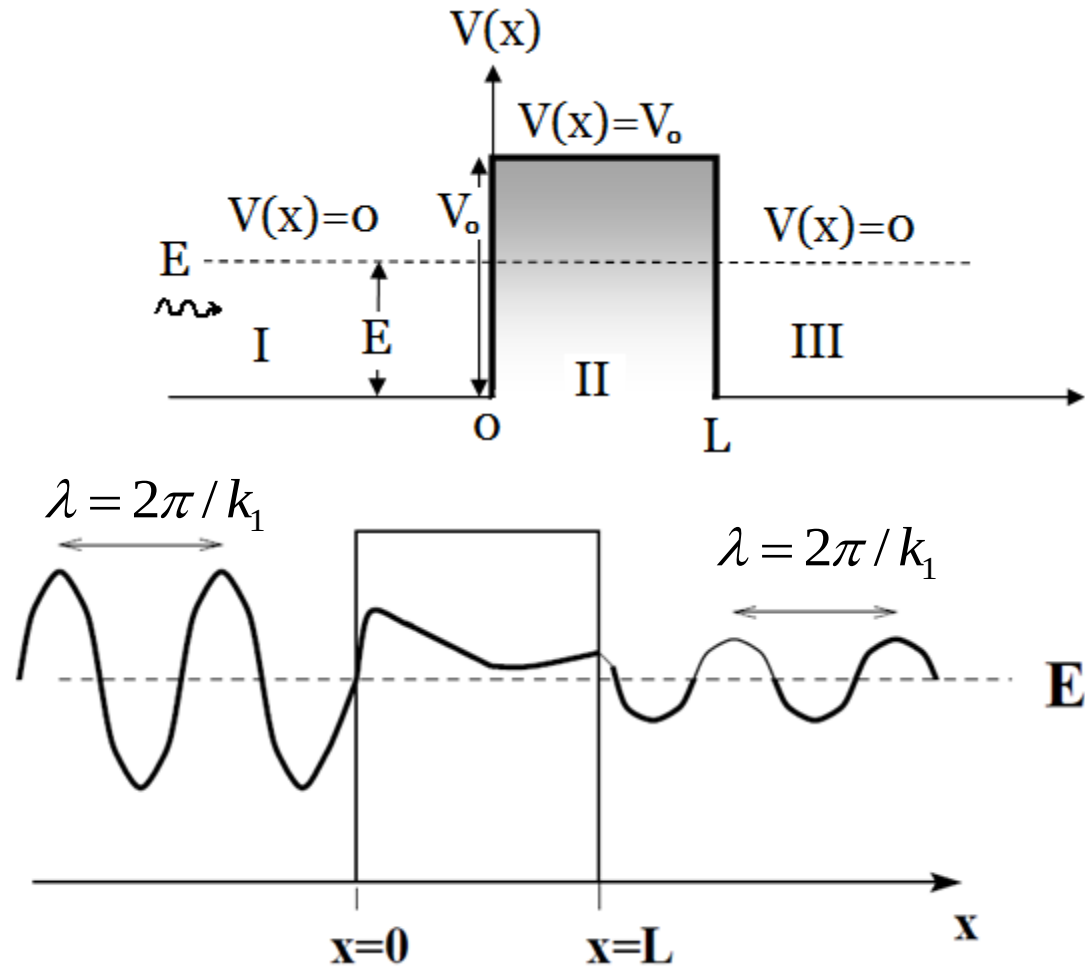
when $E < V_0$, there is a finite probability for penetration through barrier.

Higher the barrier, and the thicker it is, the smaller the chance the particle can penetrate it.



Potential Barrier

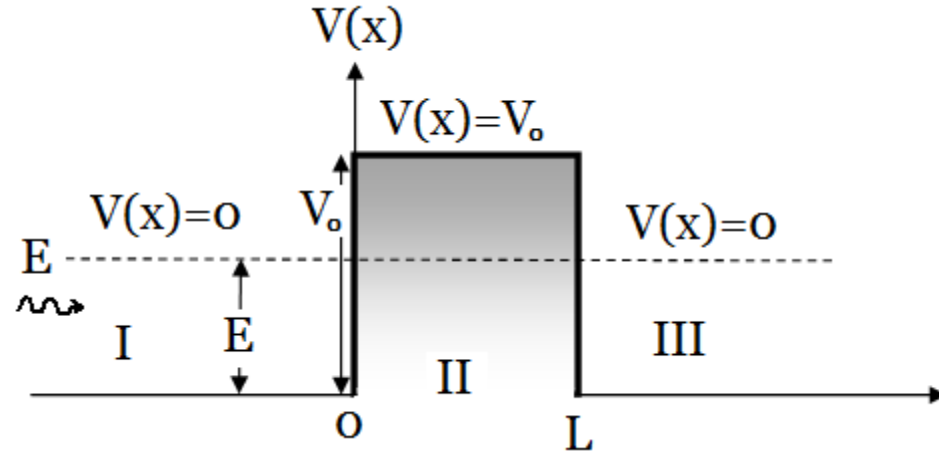
1. Can a quantum particle tunnel through a barrier with its energy less than the barrier energy?
2. How will be the amplitude of the transmitted wave as compared to the incident wave?
3. What will be The transmission probability (T)?
4. How does it depend on the width of the barrier?



Potential Barrier

Let us assume the wave particle of energy 'E' is incident to the barrier with barrier height (i.e. Barrier energy) 'V' less than E. There are three possible regions, I, II and III. Region II refers to ($x \leq 0$), the barrier region.

The Schrödinger wave equation for the region I can be expressed as below:



For region I:

$$\nabla^2 \psi_I + \frac{2mE}{\hbar^2} \psi_I = 0$$

$$\Rightarrow \nabla^2 \psi_I + k_1^2 \psi_I = 0$$

$$\Rightarrow \psi_I = Ae^{ik_1x} + Be^{-ik_1x} \dots (1)$$

For region III:

$$\nabla^2 \psi_{III} + \frac{2mE}{\hbar^2} \psi_{III} = 0$$

$$\Rightarrow \nabla^2 \psi_{III} + k_1^2 \psi_{III} = 0$$

$$\Rightarrow \psi_{III} = Fe^{ik_1x} + Ge^{-ik_1x} \dots (3)$$

For region II:

$$\nabla^2 \psi_{II} - \frac{2m(V_0 - E)}{\hbar^2} \psi_{II} = 0$$

$$\Rightarrow \nabla^2 \psi_{II} + k_2^2 \psi_{II} = 0$$

$$\Rightarrow \psi_{II} = Ce^{k_2x} + De^{-k_2x} \dots (2)$$

$$\text{where, } k_1 = \sqrt{\frac{2mE}{\hbar^2}} \text{ \& } k_2 = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

In eq. (3) $G = 0$, as there will be no reflection in region III.

Tunneling Probability

Now applying boundary conditions:

$$\text{At } x=0 \quad \psi_I = \psi_{II} \quad \& \quad \frac{d\psi_I}{dx} = \frac{d\psi_{II}}{dx}$$

$$\text{At } x=L \quad \psi_{II} = \psi_{III} \quad \& \quad \frac{d\psi_{II}}{dx} = \frac{d\psi_{III}}{dx}$$

(i) $A + B = C + D$

We will get

(ii) $ik_1(A - B) = k_2(C - D)$

(iii) $Ce^{k_2L} + De^{-k_2L} = Fe^{ik_1L}$

(iv) $k_2(Ce^{k_2L} - De^{-k_2L}) = ik_1Fe^{ik_1L}$

These four equations may be solved to get F/A .

$$k_2L \gg 1$$

We know that Transmission Probability (T) is given by:

$$T = \left(\frac{F^* F}{A^* A} \right) \left(\frac{v_{III}}{v_I} \right) = \left[\frac{16}{4 + 16(k_2 / k_1)^2} \right] e^{-2k_2L}$$

- As $L \rightarrow \infty$, $T \rightarrow 0$
- As $V_0 \gg E$, $k_2 \rightarrow \text{HIGH}$ and thus $T \rightarrow 0$
- For $V_0 \rightarrow \infty$, $T \rightarrow 0$

For low energies and wide barriers,

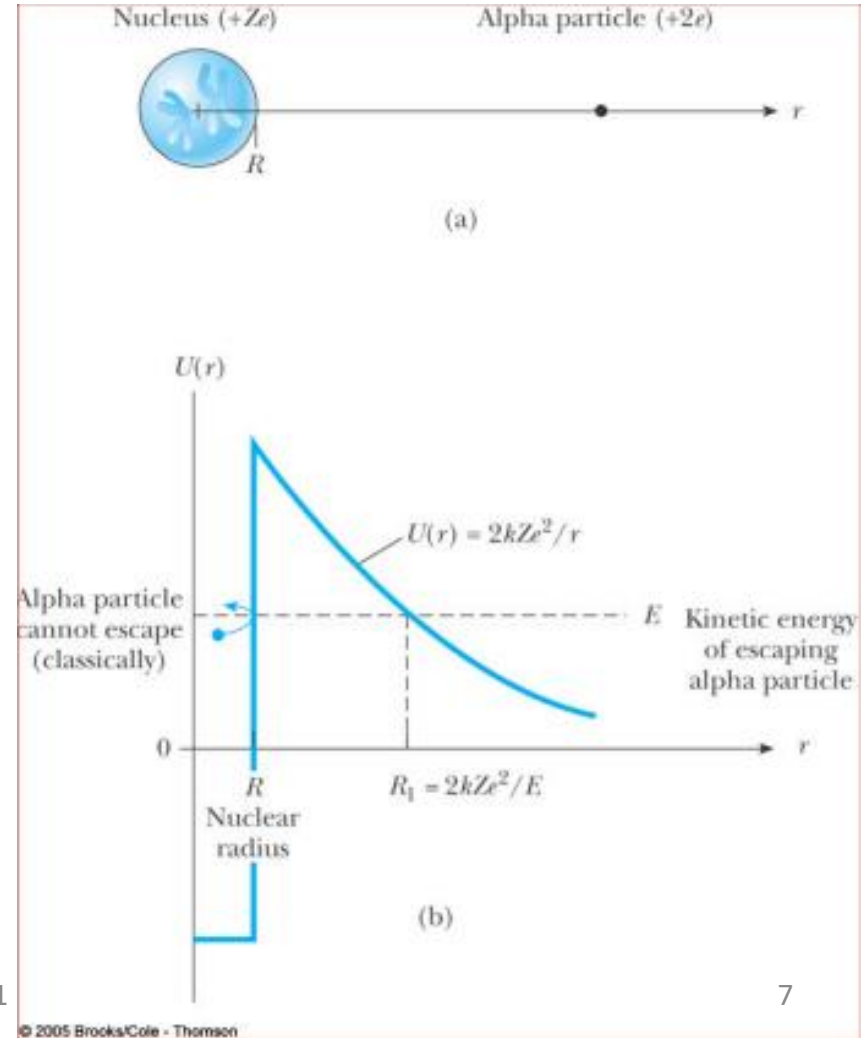
$$T \approx e^{-2k_2L} \text{ where } k_2 = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

It means the bracket quantity varies much less with E and V_0 than the exponential. The bracket quantity is always of the order of 1, So reasonable approximation of T .

Tunneling Phenomena

A particle without the energy to pass over a potential barrier may still tunnel through it.
Emission of α - particles from atomic nuclei, Penetration of particles through rectangular potential barriers, Scanning Tunneling Microscopy, Tunnel Diodes etc.

- α particles are confined in the nucleus modeled as a square well
- α particles can eventually tunnel through the Coulomb potential barrier.
- Tunneling rate is very sensitive to small changes in energy, accounting for the wide range of decay times.



Example

Po-212 atom ($Z = 84$), which presents a square barrier of width 9.4 fermi and height 26 MeV. What is the tunneling probability for an α -particle with energy 9MeV

$$T \approx e^{-2k_2L} \quad \text{where } k_2 = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$$

$$k_2 \approx 1.8 \times 10^{15} \text{ m}^{-1}$$

$$\text{Hence, } T \approx 2 \times 10^{-15}$$

Using 10^{21} attempts at the barrier per second, the number of successful attempts is about 2×10^6 per second. Therefore, the decay time for an α -particle is about $(2 \times 10^6 \text{ s}^{-1})^{-1} = 0.5 \mu\text{s}$

Example

Calculate transmission probability from given data:

If $E=1\text{eV}$, $V_0=10\text{eV}$, $L=0.50\text{nm}$ $T \approx e^{-2k_2L}$ where $k_2 = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}$

$$T \approx e^{-16} = 2.18 \times 10^{-7}$$

If $E=2\text{eV}$, $V_0=10\text{eV}$, $L=0.50\text{ nm}$ $T = 5.26 \times 10^{-7}$ (Two Times)

If barrier width is doubled $E = 1\text{eV} \rightarrow T = 4.79 \times 10^{-14}$

$$E = 2\text{eV} \rightarrow T = 2.7 \times 10^{-13}$$

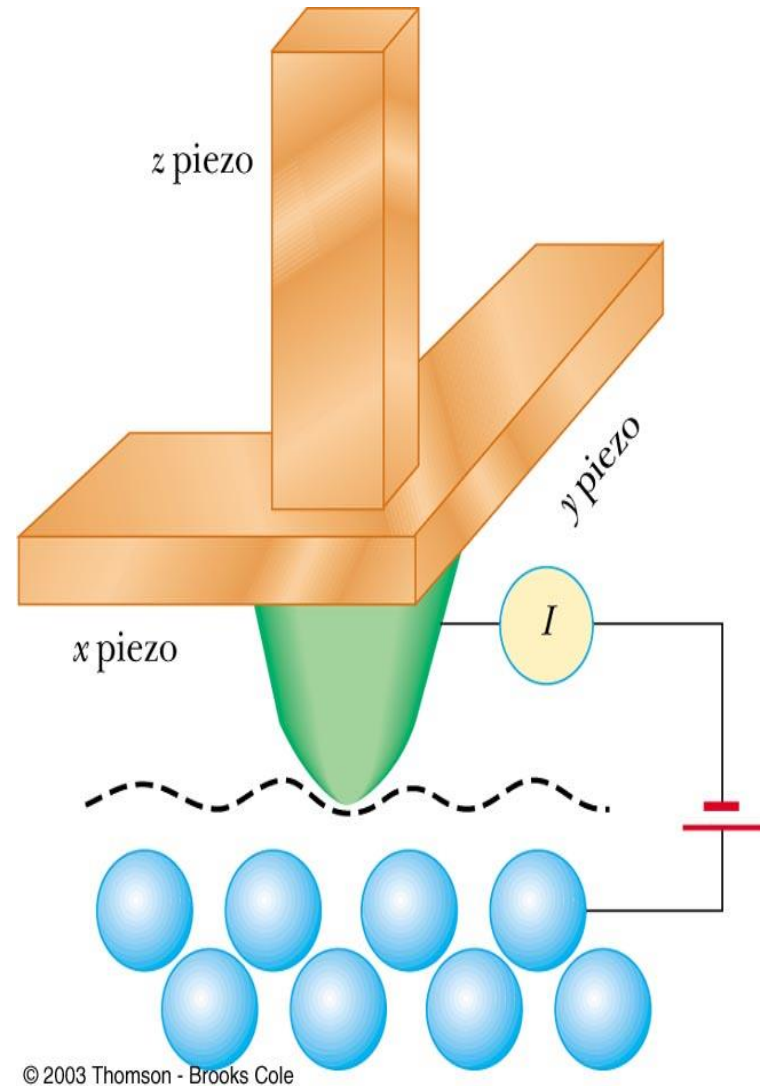
Thus, transmission probability is more sensitive to the width of the barrier than to the particle energy here.

Electrons with energies of 0.400 eV are incident on a barrier 3.00 eV high and 0.100 nm wide. Find the approximate probability for these electrons to penetrate the barrier.

A beam of electrons is incident on a barrier 6.00 eV high and 0.200 nm wide. Find the energy they should have if 1.00 percent of them are to get through the barrier.

Scanning Tunneling Microscope

- Allows highly detailed images with resolution comparable to the size of a single atom
- A conducting probe with a sharp tip is brought near the surface
- The electrons can “tunnel” across the barrier of empty space
- **By applying a voltage between the surface and the tip, the electrons can be made to tunnel preferentially from surface to tip**
- The tip samples the distribution of electrons just above the surface
- The STM is very sensitive to the distance between the surface and the tip



Allows measurements of the height of surface features within 0.001 nm

Comparison between infinite and finite potential well: Wavefunction

